# Kennesaw State University 2026-2027 Undergraduate Catalog

## Courses

Contract All Courses |

## Artificial Intelligence

- ### AI 2000: Introduction to AI Programming

**3 Credit Hours**

*Prerequisite:* CSE 1321 and CSE 1321L

**Description**

This course provides an overview of the artificial intelligence (AI) discipline and introduces the student to the fundamental principles and processes of artificial intelligence programming. This course highlights the overview of AI programming and the activities performed at each stage in the development cycle such as the introduction to AI platform (GPU) and software tools (Python, Colab), Python AI Framework (TensorFlow, PyTorch), AI data types and structures (list, matrice), AI dataset, AI ethical principles and a brief introduction to machine learning and deep learning. Through application case studies students design, implement, and test AI solutions to solve problems.

**Course Learning Outcomes**

Students who succcessfully complete this course will be able to:

- Identify, design, implement, and evaluate the AI approaches for different applications.
- Apply cutting-edge AI algorithms and tools to solve problems.
- Communicate and work effectively in a team to complete projects.
- Recognize AI ethical principles.

- ### AI 3642: Artificial Intelligence

**3 Credit Hours**

*Prerequisite:* CS 3305 or AI 2000

**Description**

The primary objective of this course is to provide a introduction to the basic principles and applications of Artificial Intelligence. It covers the basic areas of artificial intelligence including problem solving, knowledge representation, reasoning, decision making, planning, perception and action, and learning – and their applications. Students will design and implement key components of intelligent agents of modern complexity and evaluate their performance. Students are expected to develop familiarity with current research problems, research methods, and the research literature in AI.

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**Course Learning Outcomes**

Students who successfully complete this course will be able to:

1. Demonstrate knowledge of AI knowledge representation.
2. Demonstrate understanding of advanced search and reasoning algorithms.
3. Design an AI system with Lisp or Prolog for problem solving.

- ## AI 4267: Machine Learning

**3 Credit Hours**

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*Prerequisite:* AI 3642

**Description**

This course provides a broad introduction to machine learning and statistical pattern recognition including supervised, unsupervised, and ensemble learning. Topics include K-NN, Naïve Bayes Classifier, parametric and non-parametric methods, support vector machines, kernel machines, neural networks, clustering, dimensionality reduction, and model evaluation. The learning theory including bias/variance tradeoffs and large margins will be introduced. This course will also discuss recent applications of machine learning such as data mining, autonomous navigation, speech recognition, and text and web data processing.

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- ## AI 4277: Deep Learning

**3 Credit Hours**

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*Prerequisite:* AI 3642

*Concurrent:* AI 4267

**Description**

This course covers the foundations of Deep Learning; how to build neural networks and how to design successful deep learning projects. The course topics include convolutional networks, sequence modeling such as recurrent and recursive neural networks (RNNs), long short-term memory (LSTM), Adam, Dropout, BatchNorm, Xavier/He initialization, state-of-the-art technologies, and research topics leveraging Deep Learning. The course includes programming assignments in Python and in TensorFlow.

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- ## AI 4732: Machine Vision

**3 Credit Hours**

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*Prerequisite:* AI 3642

**Description**

This course introduces concepts and techniques in machine vision. Students successfully completing this course will be able to apply a variety of image processing techniques for the design and analysis of efficient algorithms for real-world applications, such as optical character recognition, face detection and recognition, motion estimation, human tracking, and gesture recognition. Topics include basic image enhancement, corner and edge detection, image morphology, linear and non-linear filters, image transformations, camera models, two-dimensional and three-dimensional image geometry, clustering and segmentation (and classification), object recognition and Bag-of-Words models, image texture, shape analysis, and tracking.

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- ## AI 4742: Natural Language Processing

**3 Credit Hours**

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*Prerequisite:* AI 3642

**Description**

This course provides an introduction to the field of natural language processing (NLP). The topics include creating systems that can understand and produce language for applications such as information extraction, machine translation, automatic summarization, question-answering, and interactive dialogue systems. This course covers linguistic (knowledge-based) and statistical approaches to language processing in the areas of syntax (language structures), semantics (language meaning), and pragmatics/discourse (the interpretation of language in context). Students will design and develop programs for analyzing and extracting information from large online corpora.

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- ## AI 4850: Applied Artificial Intelligence Capstone

**3 Credit Hours**

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*Prerequisite:* AI 3642 and completion of 90 credit hours

**Description**

The course provides a capstone experience for AI majors to promote a successful transition to the workplace or further academic study. Students will have the opportunity to practice essential project management skills and work with current AI tools and technologies. Each team will designate a team leader who is responsible for coordinating work tasks, team meetings, communications with the instructor and/or project sponsor, and team effort.

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**Course Learning Outcomes**

Students who successfully complete this course will be able to:

1. Collaborate with a team to complete a project.
2. Incorporate different elements of discipline to complete an artificial intelligence project.
3. Apply professional responsibilities based on legal and ethical principles from the AI and computing community when completing the design project.
4. Design solutions to meet project requirements by using AI algorithms and tools.
5. Effectively communicate in a variety of professional contexts including among project team and through presentations of project design.

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